

DWH + ETL (Part 1)

Why This Topic Is Critical

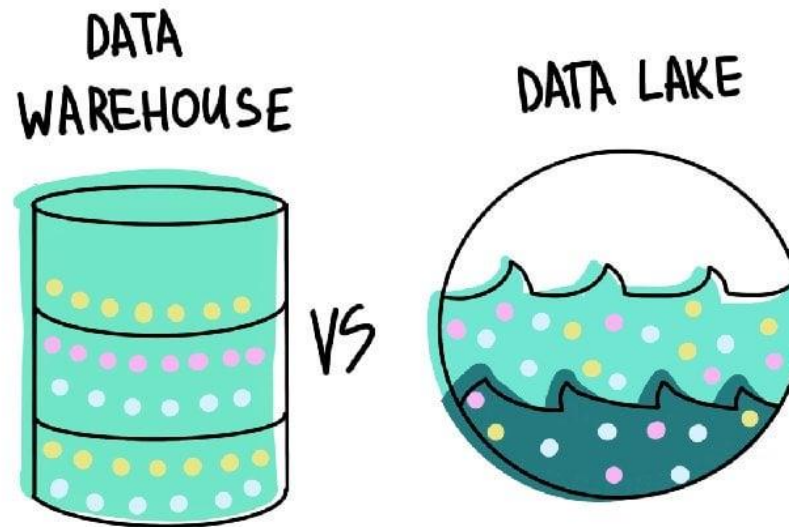
- Most BI outages begin in ingestion, not dashboards
- Small ingestion bugs create silent KPI drift
- Source systems change faster than analytics models
- Reliability requires explicit policies, not assumptions

Incident: Revenue Doubled After Rerun

- Nightly pipeline failed at 02:10 after loading 2 of 3 partitions
- Operator reran full job with plain `INSERT`
- Same transactions loaded again, revenue KPI jumped ~2x
- Root cause: no checkpoint + no idempotent target write

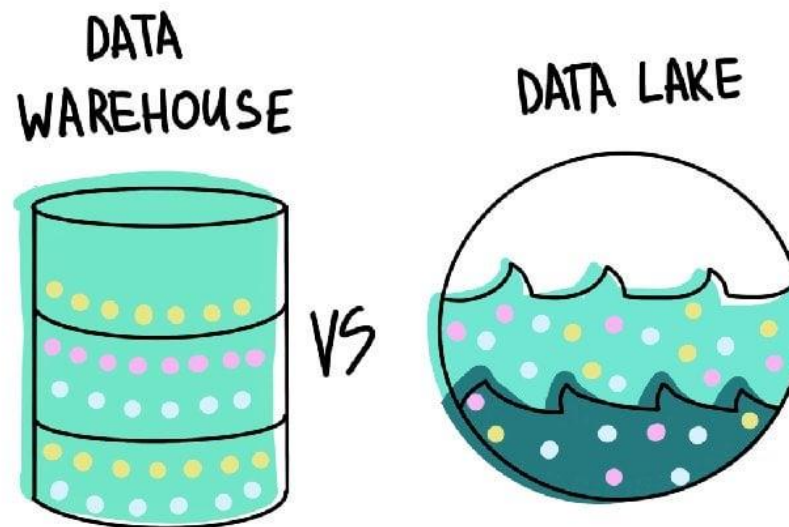
Core Stores: Warehouse vs Lake

- **Warehouse:** curated, schema-on-write, BI-first
- **Lake:** raw + processed zones, schema flexibility



Core Stores: Warehouse vs Lake

- Warehouse optimizes consistency and discoverability
- Lake optimizes ingestion agility and storage economics

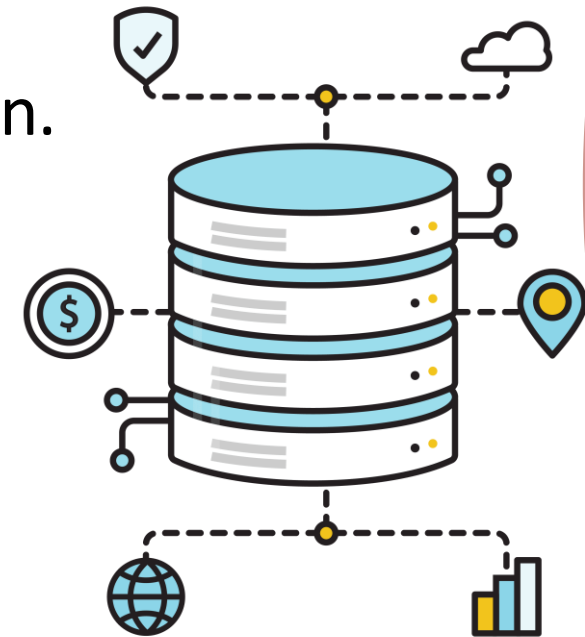


Data Warehouse Definition

- **Subject-oriented**
organized around major subjects (customer, product, sales)
- **Integrated**
multiple heterogeneous sources, cleaned and consistent
- **Time-variant**
historical perspective vs operational current-value data
- **Nonvolatile**
physically separate, operational updates do not occur in the warehouse

Why Separate Data Warehouse?

- Decision support requires historical data which operational DBs typically don't maintain.
- DS requires aggregation & summarization
- Different sources typically use inconsistent data representations



Schema-on-Write vs Schema-on-Read

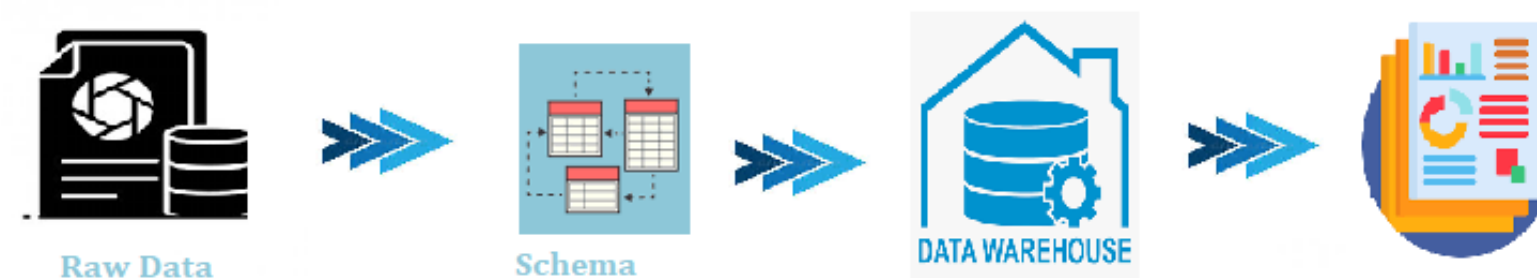
- Write-time schema validation catches issues earlier
- Read-time schema validation supports source variability
- Schema-on-write reduces BI surprise and metric drift
- Schema-on-read speeds onboarding but needs stronger governance

DWH vs Lake

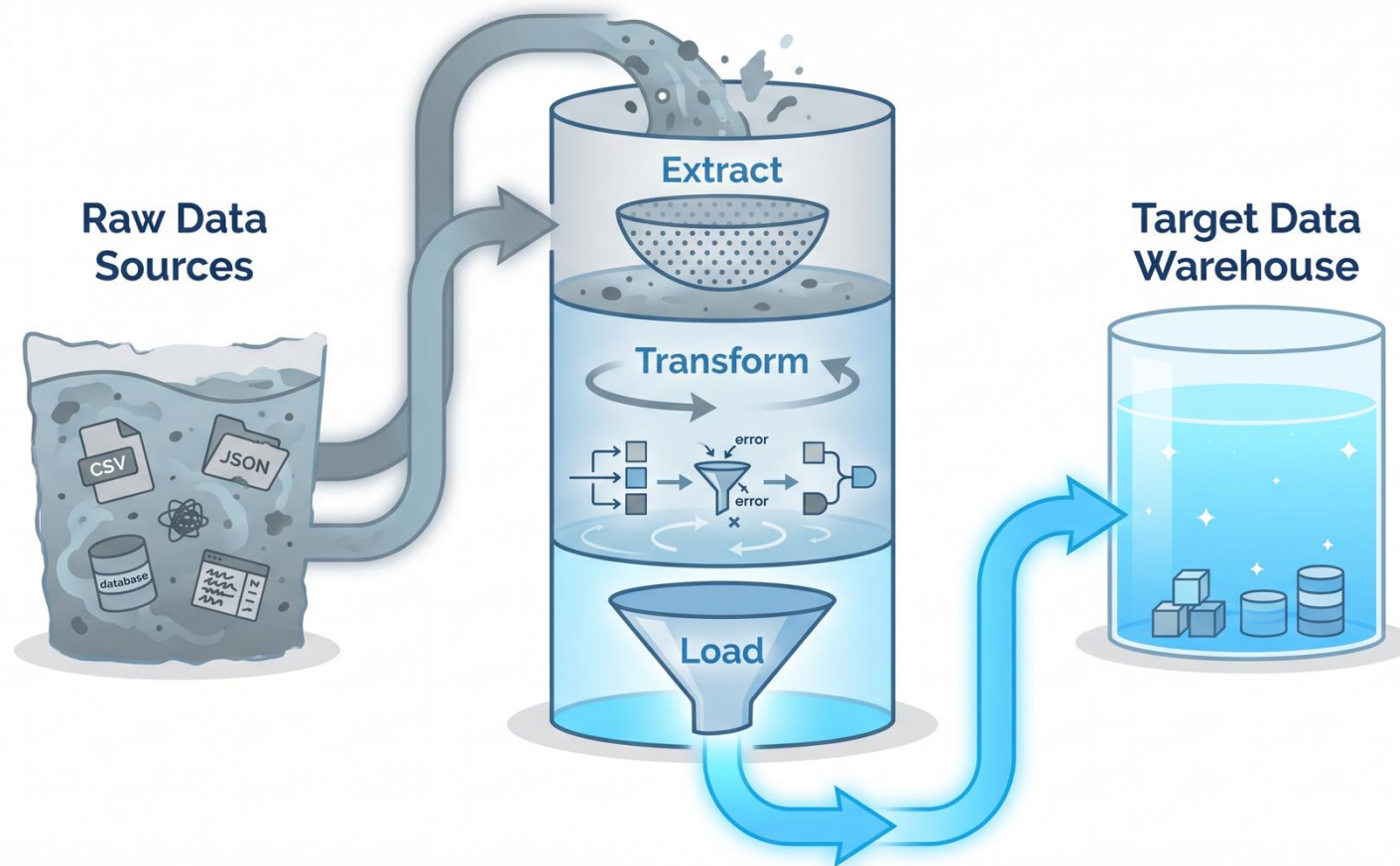
On Read



On Write



Extract, Transform, Load - ETL



Extraction

The main goal of extraction is to collect the data from the source system as fast as possible and less convenient for these source systems.



Transformation

All the collected data has been transformed into the same format

- **Filtering:** only relevant attributes going to DWH.
- **Cleaning:** Filling up the null values with specific default values.
- **Joining:** Join the multiple attributes into the one.
- **Splitting:** Splitting the single attribute into multiple attributes.
- **Sorting:** Sort the tuples based on the attributes.

Loading

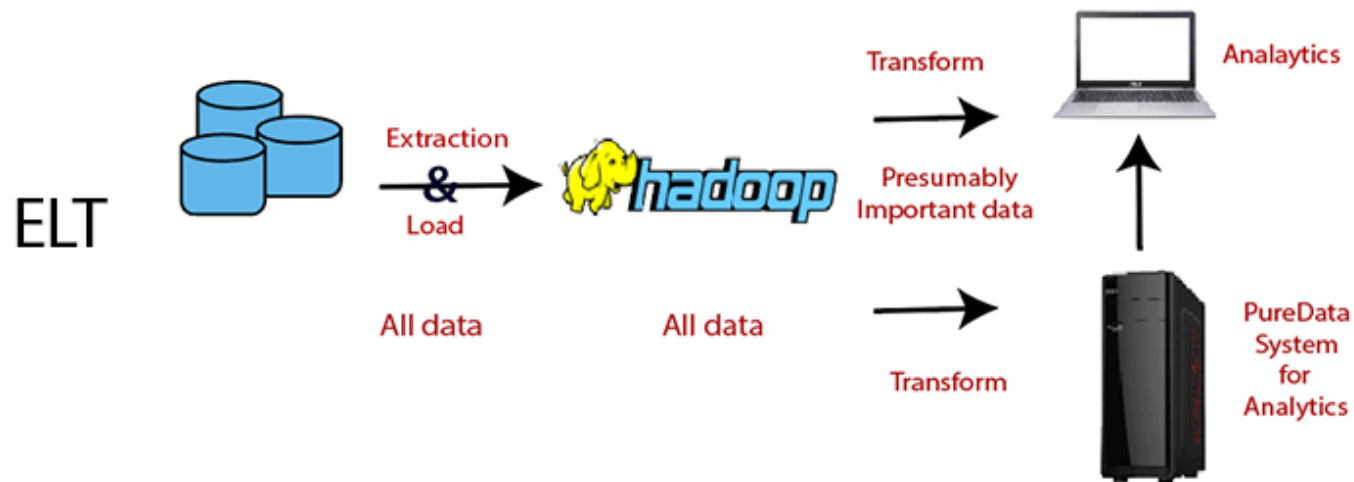
Loading is the final step of the ETL process.

The big chunk of data is collected from various sources,
transformed them,
loaded to the data warehouse.



Extract, Load, Transform - ELT

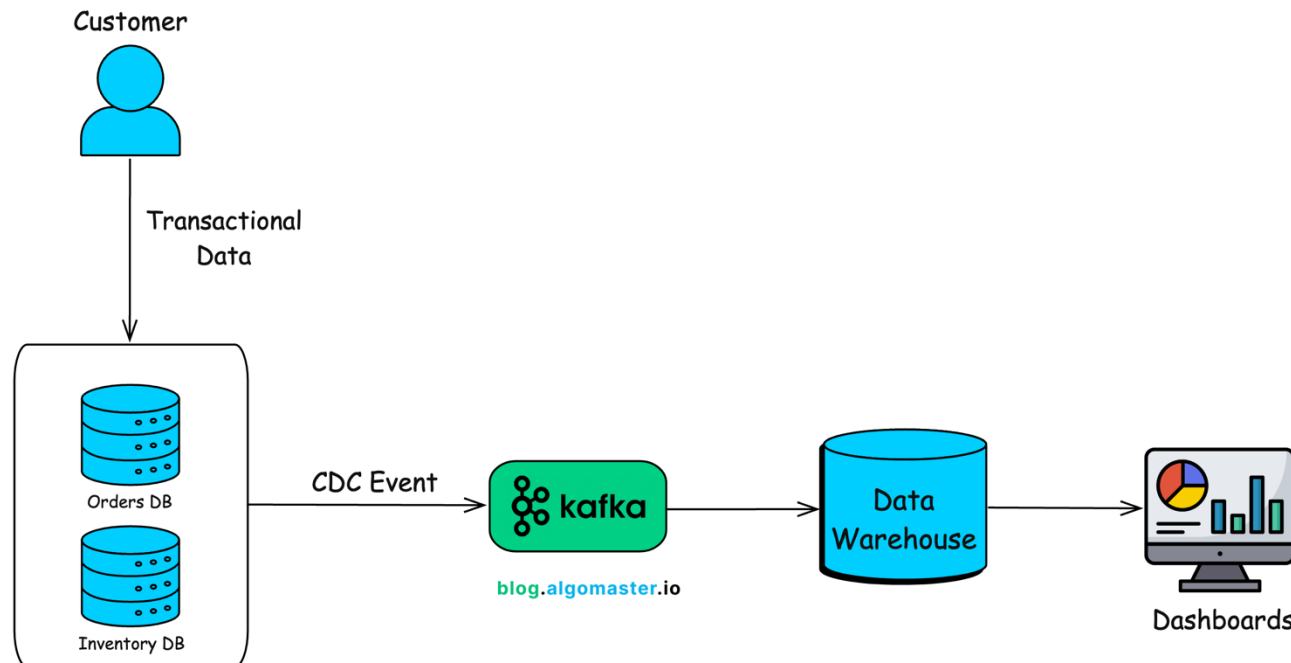
After extracting the data, we immediately start loading phase, moving all the data sources into a single, centralized data repository.



Today, most of the processes are ELT

Change-Data-Capture (CDC) Options

- **Full snapshot compare:** simple, expensive at scale
- **Timestamp:** easy, misses hard deletes unless modeled
- **Trigger-based CDC:** accurate but can burden source DB



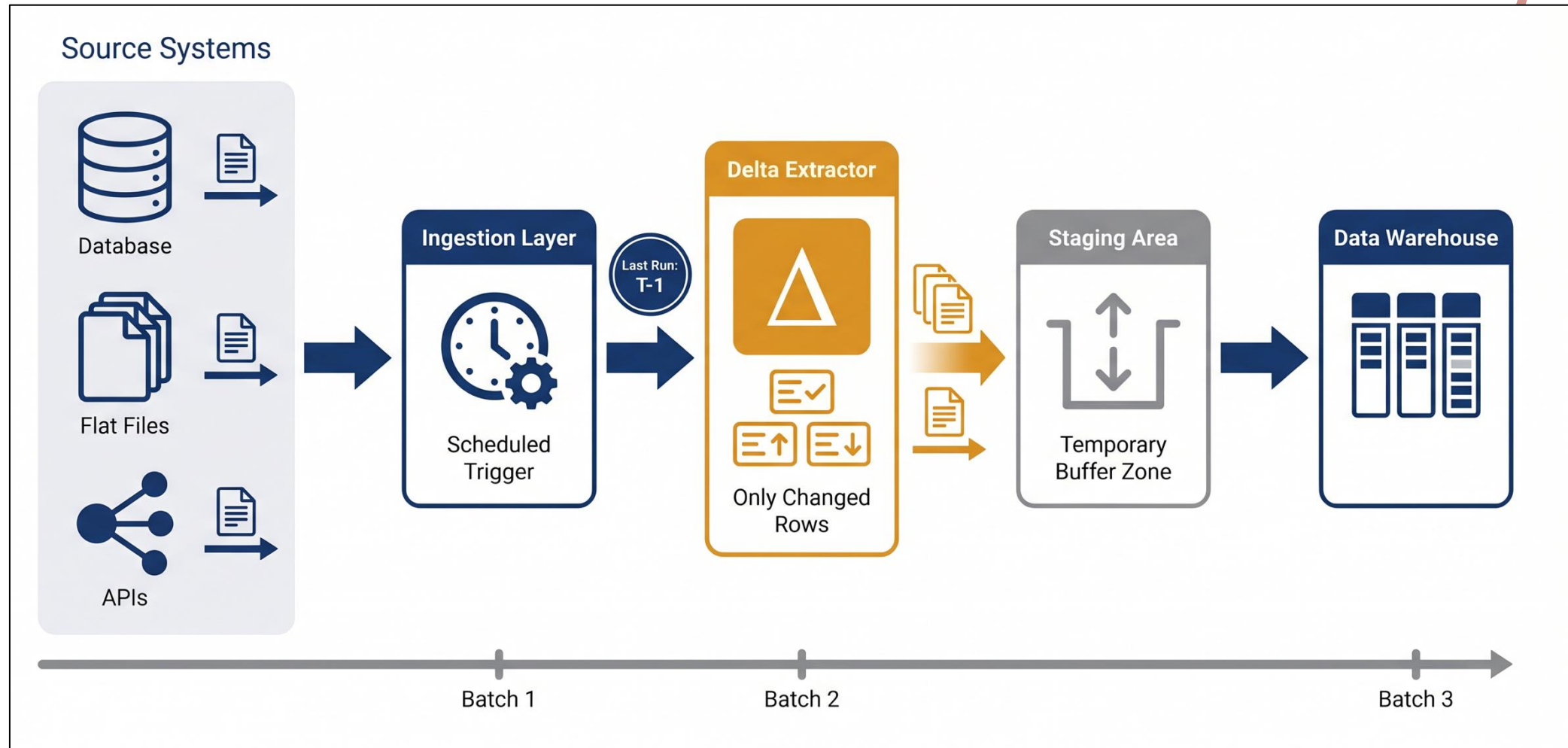
CDC Example: Orders Table

- **Inserts:** new orders must appear once
- **Updates:** status changes from `PENDING` to `PAID`
- **Deletes:** canceled rows may need tombstone semantics
- Policy must specify how each change type maps to analytics

Ingestion Modes

- Full refresh: easiest correctness, worst cost/latency at scale
- **Incremental batch: standard for most BI pipelines**
- Streaming: lowest latency, highest operational discipline

Ingestion Modes

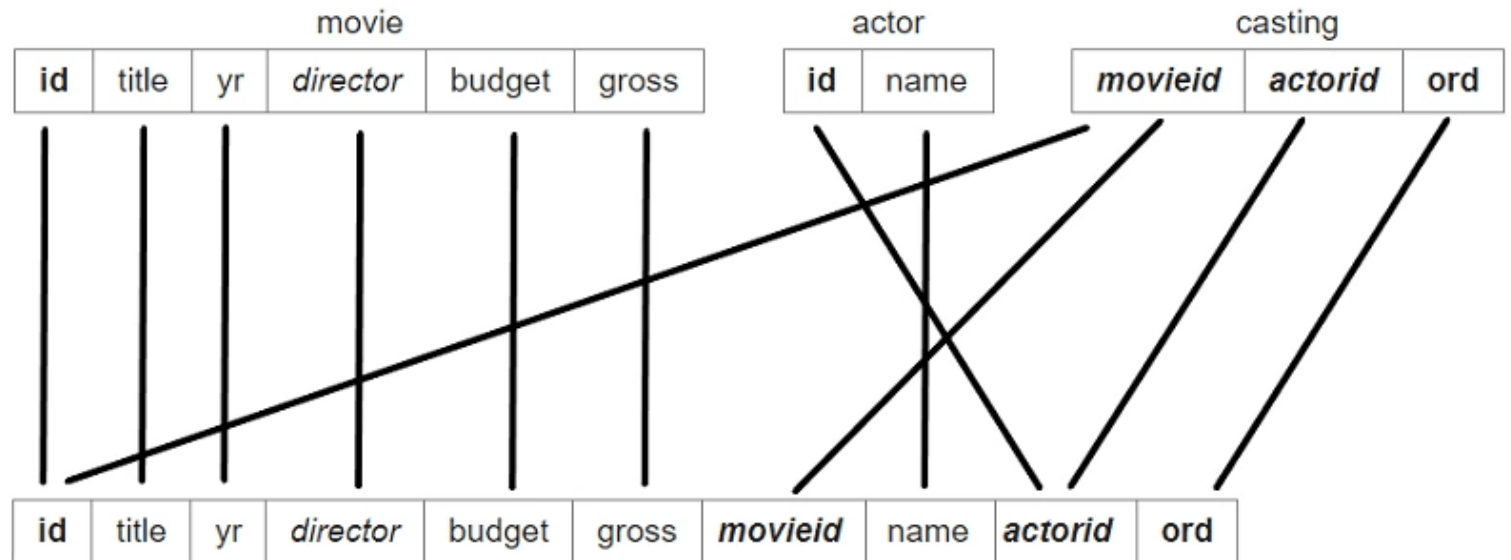


Source-to-Target Mapping (STTM)

- Source and target schemas rarely match when moving data between systems
- a set of instructions defining how to transfer content.
- Critical when integrating multiple sources into WH.

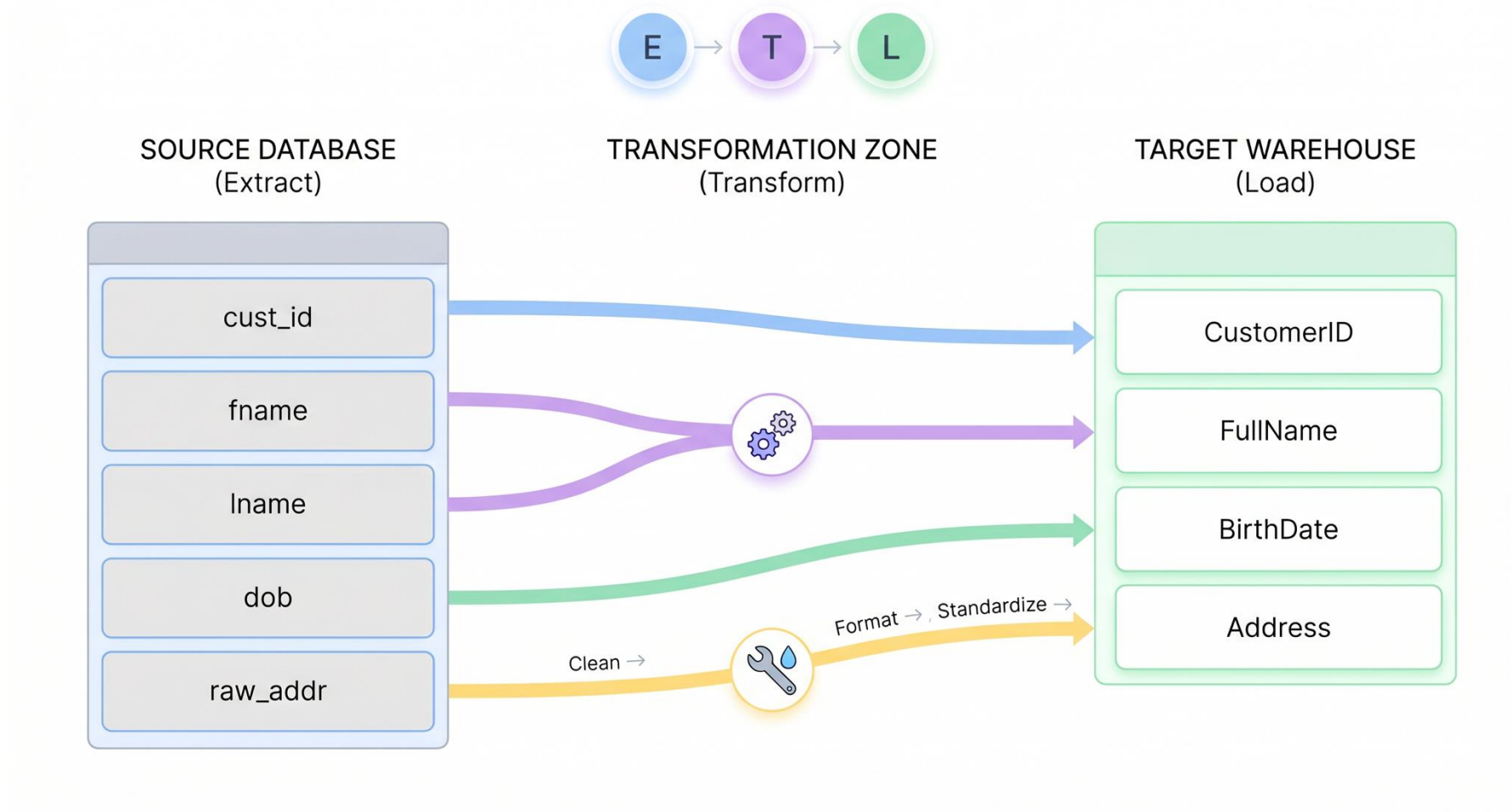
STTM Mapping Flow

Information Sources



Target Structure

STTM Process Types



Idempotency Rule

$$f(f(D)) = f(D)$$

- Reprocessing same input **cannot** change final target state.
- Requires stable keys and **deterministic** transformation rules.
- Implement with conflict-safe MERGE/upsert patterns